# The bxcjkvert Package

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**Abstract**

This package is a tailored version of the CJKvert package, accommodated for Japanese typesetting, where mixture of horizontal and vertical writings is common.

## Contents

**1 Package Loading** **1**
1.1 Options for configuring font reset . . . . . 1
1.2 Options for configuring adjustment of the baseline stretch . . . . . 1
1.3 Options for configuring initial writing direction . . . . . 2

# 1 Package Loading

`\usepackage[<option>,...]{bxcjkvert}`

The CJKvert package will be automatically loaded, if not yet loaded.
The available options are described hereafter.

## 1.1 Options for configuring font reset

When a command for switching direction (`\CJKhorz` or `\CJKvert`) is invoked, CJKvert resets the current font parameter by issuing `\normalfont`. This feature is inconvenient, particularly when authors mix different writing directions. Thus bxcjkvert suppresses the feature by default, but it can be configured by the `resetfont` option.

- `resetfont=true`: Makes direction comamnds reset the current font by issuing the `\normalfont` command.

  *Note:* This is the original behavior of CJKvert.

- `resetfont=false` (default): Makes direction comamnds retain the current font.

## 1.2 Options for configuring adjustment of the baseline stretch

CJKvert makes some adjustment to the value of `\baselinestretch` when the writing direction is changed. Namely it makes the baseline stretch enlarged by the factor of `\CJKbaselinestretch`[1] when `\CJKvert` is used.

[1] The value of `\CJKbaselinestretch` is 1.3 by default.

However as far as Japanese typesetting is concerned, there is no need to tweak the baseline stretch value. Thus bxcjkvert suppresses this feature, but again it can be configured by the `usebaselinestretch` option.

- `usebaselinestretch=true`: This is the same as the behavior of CJKvert *with* the `usebaselinestretch` option. When `\CJKvert` is invoked, the baseline stretch will be multiplied by the value of `\CJKbaselinestretch`. When `\CJKhorz` is invoked, the baseline stretch will be reverted.[2]
- `usebaselinestretch=false`: This is the same as the behavior of CJKvert *without* the `usebaselinestretch` option. When `\CJKvert` is invoked, the baseline stretch will be set to the value of `\CJKbaselinestretch`. When `\CJKhorz` is invoked, the baseline stretch will be reset to 1. This option ignores the user's setting to the baseline stretch.
- `usebaselinestretch=retain` (default): Makes direction comamnds leave the baseline stretch unchanged.

*Note:* If CJKvert is loaded with the `usebaselinestretch` option in advance, then the value of `usebaselinestretch` of this package will default to `true` instead of `retain`.

## 1.3 Options for configuring initial writing direction

CJKvert sets the initial writing direction of the document to vertical. But bxcjkvert allows authors to choose the initial direction.

- `main=true`: Sets the initial direction to vertical.

  *Note:* This is the original behavior of CJKvert.
- `main=false`: Sets the initial direction to horizontal.
- `main=retain` (default): Does not specify the initial direction. In this case, authors can use `\CJKvert` or `\CJKhorz` in the preamble to decide the initial direction.

[2]Note that, however, in this case the baseline stretch will be reverted to the value at the time when CJKvert was loaded. it might be against authors' expectation.